# Understanding Linear Discriminant Analysis

Complete Multiple-Choice Solution & Answer Key Document

**Note:** Correct answers are highlighted in green with explanations provided for each question based on fundamental machine learning principles and the theoretical limitations/assumptions of Linear Discriminant Analysis (LDA).

### 1. LDA may perform poorly when:

A) Classes are well separated

B) Data satisfies Gaussian assumptions

✓ **C) Classes have significant nonlinear boundaries**

D) Features are informative

**Explanation:** LDA projects data onto a linear decision boundary. When class boundaries are highly nonlinear, linear decision surfaces fail to capture the class structure effectively.

### 2. Which assumption of LDA is often violated in real-world datasets?

✓ **A) Equal covariance matrices across classes**

B) Presence of labels

C) Numerical features

D) Finite samples

**Explanation:** LDA assumes homoscedasticity (that all classes share the same covariance matrix $\Sigma$). In real-world data, different classes frequently exhibit different variances and feature correlations.

### 3. LDA can struggle when:

A) The number of samples is much larger than the number of features

✓ **B) The number of features exceeds the number of samples**

C) There are only two classes

D) Features are standardized

**Explanation:** When features $p >$ samples $n$, the within-class scatter matrix $S_w$ becomes singular (non-invertible), creating the classic "small sample size problem" in LDA.

### 4. Why can LDA be sensitive to outliers?

✓ **A) It uses class means and covariance estimates**

B) It performs clustering

C) It ignores feature values

D) It uses decision trees

**Explanation:** Sample mean vectors and sample covariance matrices are non-robust statistics that are heavily distorted by extreme values/outliers.

**5. Which of the following is a limitation of LDA?**

A) Requires labeled data

B) Assumes normally distributed classes

C) Assumes equal class covariances

✓ **D) All of the above**

**Explanation:** LDA is a supervised algorithm (requires labels), relies on Gaussian distributions, and assumes equal class covariance matrices. Therefore, all listed points are key limitations.

**6. When class covariance matrices are substantially different, which method may outperform LDA?**

✓ **A) Quadratic Discriminant Analysis (QDA)**

B) PCA

C) K-Means

D) KNN Regression

**Explanation:** QDA relaxes the equal covariance assumption of LDA and estimates individual covariance matrices for each class, resulting in quadratic decision boundaries.

**7. LDA assumes that class boundaries are:**

✓ **A) Linear**

B) Circular

C) Arbitrary

D) Fractal

**Explanation:** LDA derives linear combinations of features to separate classes, leading to linear decision hyperplanes.

**8. What problem arises when the within-class scatter matrix S_w is singular?**

A) Projection becomes easier

✓ **B) Matrix inversion becomes difficult or impossible**

C) Classes become balanced

D) Dimensionality increases automatically

**Explanation:** Finding the optimal linear projection in LDA requires computing $S_w^{-1}$. If S_w is singular, its inverse does not exist without regularization (e.g., Regularized LDA / Ridge penalty).

**9. LDA may not effectively separate classes when:**

A) The classes are linearly separable

✓ **B) The classes overlap heavily**

C) Features are relevant

D) Labels are correct

**Explanation:** Heavy overlap in feature distributions between classes reduces separability and increases classification error rate regardless of the linear hyperplane chosen.

**10. In high-dimensional datasets with few samples, LDA often suffers from:**

✓ **A) Overfitting and unstable covariance estimates**

B) Underfitting only

C) Reduced computation time

D) Automatic feature selection

**Explanation:** High-dimensional settings with small sample sizes lead to high-variance/unstable estimations of the covariance matrix and severe overfitting.

**11. Which data characteristic can negatively impact LDA performance?**

A) Strong outliers

B) Noisy measurements

C) Non-Gaussian distributions

✓ **D) All of the above**

**Explanation:** Outliers skew estimates, noise degrades boundary precision, and non-Gaussian distribution violates core model assumptions. All three factors negatively affect performance.

**12. LDA is generally less suitable than kernel-based methods when:**

A) Data is linearly separable

✓ **B) Data contains nonlinear class structures**

C) Data has few features

D) There are only two classes

**Explanation:** Kernel-based methods (like Kernel LDA or Support Vector Machines with RBF kernels) project data into higher-dimensional feature spaces to handle complex nonlinear relationships that standard linear LDA cannot model.

**13. A major limitation of LDA compared to PCA is that:**

✓ **A) It requires class labels**

B) It can handle more dimensions

C) It is unsupervised

D) It ignores covariance

**Explanation:** PCA is an unsupervised technique that works without labels, whereas LDA requires explicit target class labels to compute between-class and within-class scatter matrices.

**14. Why might class imbalance affect LDA?**

✓ **A) Larger classes can dominate mean and covariance estimation**

B) LDA automatically balances classes

C) LDA ignores sample counts

D) Class imbalance improves separability

**Explanation:** In pooled covariance matrix calculation, majority classes carry greater weight, shifting the estimated mean vectors and boundary toward favoring the dominant class.

**15. For a dataset with C classes, a limitation of LDA is that the maximum number of discriminant components is:**

A) Equal to the number of samples

B) Equal to the number of features

✓ **C) C - 1**

D) Unlimited

**Explanation:** The between-class scatter matrix $S_b$ has a rank of at most C - 1 (since the C class means sum to the global mean), bounding the number of useful non-zero eigenvalues/components to min(p, C - 1).

**16. Which statement best describes a limitation of classical LDA?**

✓ **A) It assumes a common covariance structure across all classes**

B) It works perfectly for any nonlinear dataset

C) It does not require labels

D) It is immune to outliers

**Explanation:** Classical LDA strictly assumes equal covariance structures (homoscedasticity) across all target classes.

**17. LDA may produce suboptimal results when:**

✓ **A) Feature distributions deviate significantly from normality**

B) Classes are clearly separated

C) Data is low-dimensional

D) Covariances are equal

**Explanation:** LDA relies on maximum likelihood estimation assuming Gaussian distributions; significant non-normality degrades decision rule optimality.

**18. Which of the following is known as the "small sample size problem" in LDA?**

A) Too many classes compared to features

✓ **B) Too many features compared to samples**

C) Too many samples compared to features

D) Too few labels

**Explanation:** When the feature dimension p exceeds the sample size n, $S_w$ becomes singular and non-invertible, known as the small sample size (SSS) problem.

**19. LDA cannot extract more than C - 1 discriminant features because:**

✓ **A) The rank of the between-class scatter matrix is limited**

B) NumPy restricts dimensions

C) Eigenvalues are always zero

D) PCA removes extra dimensions

**Explanation:** The maximum rank of $S_b$ (the between-class scatter matrix constructed from C class mean vectors) is C - 1, which caps the number of non-zero eigenvectors/components.

**20. Which scenario is most likely to violate LDA assumptions?**

A) Two Gaussian classes with equal covariance matrices

✓ **B) Multiple classes with highly nonlinear boundaries and unequal covariances**

C) Balanced classes with low noise

D) Well-separated classes in low dimensions

**Explanation:** Having nonlinear class boundaries and unequal covariances violates both key fundamental assumptions of LDA simultaneously (linear separability and homoscedasticity).